

Notes: Antill & Hunter (JF, 2026)
Consumer Choice and Corporate Bankruptcy

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1 Big picture

- **Question.** How large are the indirect costs of corporate bankruptcy that arise because customers avoid bankrupt firms?
- **Core challenge.** In observational data, firms file for bankruptcy after negative shocks. Those shocks also reduce demand, so demand losses cannot be attributed cleanly to the bankruptcy filing itself.
- **Design.** The paper uses four incentivized survey experiments. The first three are multiple-price-list experiments that randomly reveal real Chapter 11 filings and elicit willingness to pay. The fourth is a mechanism experiment that varies hypothetical bankruptcy information across airlines, retail firms, and car manufacturers.
- **Main result.** Learning that a firm is in Chapter 11 reduces willingness to pay by about 17% to 28% in the price-list experiments and about 18% to 24% in the mechanism experiment.
- **Awareness.** Consumers are not generally aware of prebankruptcy financial distress, but many are aware once major firms actually file for bankruptcy. On average, about 38% of consumers know about major consumer-facing bankruptcies.
- **Mechanism.** Consumers mainly worry that bankruptcy can reduce current product/service quality and can prevent valuable future interactions, such as warranty service, returns, reward programs, and continued relationship value.
- **Structural implication.** Combining experimental estimates with airline market-share and price data, the paper estimates that customer-attrition costs are about 12% to 15% of enterprise value.
- **Corporate-finance takeaway.** Indirect bankruptcy costs are not just theoretical residuals. Consumer behavior can generate large value losses in Chapter 11, helping rationalize why expected default costs matter for capital-structure decisions.

2 Incremental contribution of the paper

- **Direct causal evidence on consumer demand.** Prior work inferred distress costs from market outcomes that confound bankruptcy with bad fundamentals. This paper uses randomized information provision to isolate the causal effect of the bankruptcy label on willingness to pay.
- **Experimental corporate-finance methodology.** The paper imports methods from marketing and experimental economics into corporate finance: BDM-style price lists, randomized bankruptcy information, awareness measurement, and incentive-compatible willingness-to-pay elicitation.
- **Awareness measurement.** The paper is the first to directly measure how many consumers know about major corporate bankruptcies and to show that awareness is substantial after filing but weak before filing.
- **Mechanism decomposition.** The fourth experiment separates three mechanisms: current-quality concerns, future-interaction concerns, and negative quality-signaling concerns. It shows that current quality and future interactions matter most.
- **Link from experiments to dollar costs.** The paper does not stop at WTP effects. It embeds the experimental estimates into a structural differentiated-products model to estimate market-share, price, profit, and consumer-surplus consequences of historical airline bankruptcies.
- **New interpretation of bankruptcy costs.** The paper shows that customer attrition alone can account for a large part of the bankruptcy-cost magnitudes discussed in the corporate-finance literature.
- **Practical implication.** Because consumers often underestimate bankruptcy survival probabilities, firms, courts, or regulators may be able to reduce indirect costs by communicating accurate information about Chapter 11 continuity and survival prospects.

3 Data and measurement

3.1 Price-list experiments: Hertz, Instant Brands, and Spirit Airlines

The first three experiments focus on firms that were actually in Chapter 11 at the time of the survey.

- **Hertz experiment.** Conducted during Hertz’s bankruptcy. Participants choose between a \$50 Hertz gift card and Enterprise gift cards of varying value. Treatment-group participants are told that Hertz filed for Chapter 11 and remains in bankruptcy.
- **Instant Brands experiment.** Conducted after Instant Brands filed for bankruptcy. Participants read a news-style article; bankruptcy is made less salient in the treatment version. The prize is an Instant Pot, so the design addresses concerns that gift cards might not be honored.
- **Spirit Airlines experiment.** Conducted during Spirit’s bankruptcy and uses a \$50 Spirit Airlines gift card. The larger sample makes the first-stage and WTP effects more precise.

Interpretation: The design holds the firm and product fixed while randomly changing whether participants learn about bankruptcy. This is what observational studies of bankrupt firms cannot do.

3.2 Awareness and willingness-to-pay measures

In each price-list experiment, the paper constructs:

- $Treat_i$: randomized assignment to receive bankruptcy information.
- $Aware_i$: whether participant i correctly identifies the firm as bankrupt after the price list.
- WTP_i : the participant’s willingness to pay for the bankrupt firm’s product or gift card.
- X_i : controls for prior purchasing from the focal firm and rivals, demographics, and misconceptions about Chapter 11 survival.

Interpretation: Treatment is an instrument for awareness because some control participants already know about the bankruptcy and some treatment participants may not update fully.

3.3 Mechanism experiment

The mechanism experiment studies airlines, retail firms, and car manufacturers using hypothetical purchase decisions. Participants answer 10 WTP questions in one of seven randomized information groups:

1. control;
2. bankruptcy;
3. quality reassurance;
4. survival 50;
5. survival 100;
6. prebankruptcy;
7. postbankruptcy.

Each participant remains in the same information group across all questions.

Interpretation: The mechanism experiment can vary bankruptcy states for real and fictional firms without deception because decisions are explicitly hypothetical, but participants still receive prizes based on the preferences implied by their choices.

3.4 Awareness survey

The paper separately surveys consumers about 99 firms: 70 previously bankrupt firms and 29 never-bankrupt firms. For each firm, consumers report whether they think the firm has filed for bankruptcy and whether a hypothetical bankruptcy would reduce their willingness to pay.

Interpretation: This survey answers an external-validity question: experimental WTP effects matter only if consumers actually know about bankruptcies outside the experiment.

3.5 Structural airline data

For model-based cost estimates, the paper focuses on airlines because airline routes have rich price and market-share data. It uses the Department of Transportation DB1B data for markets around Delta, American, and United Airlines bankruptcies. The model estimates taste parameters from experiments and awareness surveys, then fits airline market shares and prices to recover route-airline tastes and marginal costs.

4 Empirical specifications and model

4.1 Price-list first stage and 2SLS

The first stage is:

$$Aware_i = \pi_0 + \pi_1 Treat_i + X_i' \gamma + u_i. \quad (1)$$

The second stage is:

$$WTP_i = \alpha + \beta \widehat{Aware}_i + X_i' \delta + \varepsilon_i. \quad (2)$$

Interpretation: The coefficient β is the causal effect of becoming aware that the firm is bankrupt for compliers: participants whose awareness is moved by the experimental information.

4.2 Mechanism experiment WTP equation

For participant i and question q , the normalized WTP is:

$$WTP_{iq}^{norm} = \frac{\text{WTP at Firm B}}{\text{Given price at Firm A}}. \quad (3)$$

The estimated equation is:

$$WTP_{iq}^{norm} = \alpha + \kappa Prebankruptcy_i + \beta Bankruptcy_i + \eta Postbankruptcy_i \quad (4)$$

$$+ \delta Quality_i + \gamma Survival50_i + \rho Survival100_i + \varepsilon_{iq}. \quad (5)$$

Interpretation: The omitted category is the control group. The coefficient on Bankruptcy measures the baseline WTP penalty from being told the firm is in Chapter 11. Quality and survival treatments reveal which concerns drive the penalty.

4.3 Awareness and firm-characteristics regressions

The paper estimates firm-level regressions such as:

$$Awareness_f = \alpha + \beta Bankruptcy_f + Z_f' \theta + \xi_f, \quad (6)$$

and analogous regressions for the share of consumers who would reduce WTP if the firm filed.

Interpretation: These regressions quantify which firms consumers remember and which industries create the strongest negative reactions.

4.4 Structural discrete-choice model

For airline j in market t , consumer i receives indirect utility:

$$u_{ijt} = \delta_{jt} + \alpha_i p_{jt} + \beta_i A_{ijt} B_{jt} + \beta_i^{Post} A_{ijt}^{Post} B_{jt}^{Post} + \varepsilon_{ijt}. \quad (7)$$

Here B_{jt} indicates current bankruptcy and B_{jt}^{Post} indicates past bankruptcy. Awareness indicators determine whether bankruptcy status affects utility.

Consumers choose the alternative with the highest utility, including an outside option. Market shares take a logit form:

$$S_{jt}^{model}(p_t) = E \left[\frac{\exp(\delta_{jt} + \alpha_i p_{jt} + \beta_i A_{ijt} B_{jt} + \beta_i^{Post} A_{ijt}^{Post} B_{jt}^{Post})}{1 + \sum_k \exp(\delta_{kt} + \alpha_i p_{kt} + \beta_i A_{ikt} B_{kt} + \beta_i^{Post} A_{ikt}^{Post} B_{kt}^{Post})} \right]. \quad (8)$$

Airline j chooses price to maximize:

$$\max_{p_{jt}} S_{jt}^{model}(p_{jt}, p_{-j,t}^*) (p_{jt} - c_{jt}). \quad (9)$$

Interpretation: The model translates WTP penalties into equilibrium effects on market share, prices, consumer surplus, and profits.

5 Identification strategy

5.1 Experimental identification

- Random treatment assignment creates exogenous variation in bankruptcy awareness.
- Incentive-compatible BDM price lists encourage truthful reporting of WTP.
- The 2SLS design handles imperfect compliance because many control participants are already aware of some bankruptcies.
- The mechanism experiment holds product and firm details fixed while varying bankruptcy status and details of the bankruptcy.

5.2 Addressing experimenter demand

The paper tests experimenter demand by measuring whether participants believe the survey is about bankruptcy. Table II shows that the estimated WTP penalty remains large even when participants do not think the survey purpose relates to bankruptcy.

5.3 External validity

The awareness survey shows that many consumers outside the experiment know about major corporate bankruptcies. Subsample tests in Table VIII show that likely-aware consumers react similarly to likely-unaware consumers in most experiments.

5.4 Structural identification

Experimental variation identifies price sensitivity and bankruptcy sensitivity. The awareness survey identifies bankruptcy awareness rates and forgetting. DB1B route data identify average airline-market tastes and marginal costs by matching model-implied and observed prices and market shares.

5.5 Limits

The structural estimates isolate costs due to lost customers and omit benefits of Chapter 11, such as contract rejection, DIP financing, and reduced debt overhang. They should therefore be interpreted as customer-attrition costs, not net welfare effects of bankruptcy.

6 Tables and interpretations

6.1 Table I: Price List Experiments

Read: Treatment strongly increases awareness in all three experiments: 0.677 for Hertz, 0.898 for Instant Brands, and 0.565 for Spirit. Instrumented awareness reduces WTP by \$13.716 for Hertz, \$10.778 for Instant Pot, and \$8.762 for Spirit. Relative to control means, these effects are about 28%, 20%, and 17%.

Interpretation: The table provides the core causal estimates. The first stage is strong and the second-stage WTP effects are economically large.

Economic message: Bankruptcy directly lowers demand for consumer products, even when the product, firm, and price list are otherwise unchanged.

Table I
Price List Experiments

This table presents the results of our three price list experiments (Section I). In each experiment, $Aware_i$ is an indicator equal to one for participants i who are aware of the relevant firm's bankruptcy (Hertz in experiment one, Instant Brands in experiment two, Spirit Airlines in experiment three). $Treat_i$ is an indicator for i being randomly selected to learn that the firm is bankrupt. WTP_i denotes the willingness to pay for the experiment prize (a \$50 Hertz gift card in experiment one, an Instant Pot in experiment two, a \$50 Spirit Airlines gift card in experiment three). Columns (1), (3), and (5) report results of ordinary least squares (OLS) regressions of $Aware_i$ on $Treat_i$ for the first, second, and third experiment samples, respectively. Columns (2), (4), and (6) report results of 2SLS regressions of WTP_i on $Aware_i$, where we instrument for awareness using $Treat_i$, for the first, second, and third experiment samples, respectively. We report the F -statistic for $Treat_i$ in columns (1), (3), and (5). We report the mean of the dependent variable, calculated using only control-group participants, in each column. All regressions include the control variables described in Section I. We report robust standard errors in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

	Aware (1)	WTP (2)	Aware (3)	WTP (4)	Aware (5)	WTP (6)
Treat	0.677*** (0.043)		0.898*** (0.039)		0.565*** (0.020)	
Aware of bankruptcy		-13.716*** (3.456)		-10.778** (5.474)		-8.762*** (2.369)
Experiment Estimator	Hertz OLS	Hertz 2SLS	Instant OLS	Instant 2SLS	Spirit OLS	Spirit 2SLS
Control variables	Y	Y	Y	Y	Y	Y
Observations	356	356	161	161	1,551	1,551
F -statistic	251.1		523.4		765.4	
Control-group mean	0.230	49.20	0.0200	53.48	0.330	50.63

Table II
Experimenter Demand

This table presents 2SLS regressions that reveal the extent to which experimenter demand drives our results. Each column presents the results of a 2SLS regression of WTP_i on $Aware_i$, where we instrument for awareness using $Treat_i$ (see Table I for details). In column (1), the sample includes only those participants in our Spirit Airlines experiment that believe the survey's purpose *does not* relate to bankruptcy. In column (2), the sample includes only those participants in our Spirit Airlines experiment that believe the survey's purpose *does* relate to bankruptcy. In this sense, experimenter demand is more relevant in column (2) than column (1). In column (3), the sample includes all participants from our Hertz experiment; this replicates column (2) of Table I. We replicate our Hertz experiment in an "unincentivized" experiment in which participants do not receive any prizes (gift card choices are hypothetical). In column (4), the sample includes all participants from this additional unincentivized Hertz study. In this sense, experimenter demand is more relevant in column (4) (no incentives) than in column (3) (incentives). All regressions include the control variables described in Section I. We report robust standard errors in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

	WTP			
	(1)	(2)	(3)	(4)
Aware	-8.906*** (2.698)	-10.682* (6.132)	-13.716*** (3.456)	-9.344*** (2.579)
Experiment	Spirit Low Y	Spirit High Y	Hertz Low Y	Hertz High Y
Experimenter demand				
Control variables	Y	Y	Y	Y
Observations	1,046	505	356	406

6.2 Table II: Experimenter Demand

Read: The WTP penalty remains large in subsamples where participants are less likely to believe the survey is about bankruptcy. The low-demand Spirit estimate is -8.906, close to the main effect.

Interpretation: The result reduces concern that participants simply say they dislike bankrupt firms because they think the experimenter expects that answer.

Economic message: The WTP penalty is not an artifact of demand effects; consumers genuinely dislike the bankruptcy information.

6.3 Table III: Mechanism Experiment Setup: Information Groups

Read: The table lists the seven randomized information conditions: control, bankruptcy, quality reassurance, survival 50, survival 100, prebankruptcy, and postbankruptcy.

Interpretation: This table is the roadmap for the mechanism experiment. Each treatment isolates a different interpretation of the bankruptcy signal.

Economic message: The design separates demand losses caused by current-quality concerns, survival/future-interaction concerns, and longer-run reputation effects.

Table III
Mechanism Experiment Setup: Information Groups

Each participant is randomized into one of seven information groups. Once a participant is assigned to an information group, they see the following text describing Firm B in all 10 willingness-to-pay questions. See [Internet Appendix Table IAX](#) for the identity of Firm B in each question. The exact text in the quality treatment varies across industries.

Group	Example Text
Control	
Bankruptcy	Please imagine that [Firm B] filed for Chapter 11 bankruptcy and is still in bankruptcy.
Quality	Please imagine that [Firm B] filed for Chapter 11 bankruptcy and is still in bankruptcy. During the bankruptcy, the Better Business Bureau assessed that [Firm B's] quality was not affected by the bankruptcy.
Survival 50	Please imagine that [Firm B] filed for Chapter 11 bankruptcy and is still in bankruptcy. Financial experts estimate that there is a 50% chance that [Firm B] will emerge from bankruptcy and continue operating.
Survival 100	Please imagine that [Firm B] filed for Chapter 11 bankruptcy and is still in bankruptcy. Financial experts estimate that [Firm B] will almost certainly emerge from bankruptcy and continue operating.
Prebankruptcy	Please imagine that financial experts estimate that [Firm B] has a 50% chance of filing for Chapter 11 bankruptcy in the next six months.
Postbankruptcy	Please imagine that [Firm B] filed for Chapter 11 bankruptcy, emerged, and is now operating as a nonbankrupt company.

Table IV
How, When, and Why does Bankruptcy Change Consumers' Willingness to Pay?

This table presents OLS regressions of willingness to pay on indicators for treatment groups. For each participant in each of 10 questions, we measure the participant's willingness to pay for some Firm B's product or service. In each question, we tell the participant how much an equivalent product costs at another Firm A. We define the normalized willingness to pay as the ratio of the willingness to pay for Firm B's product to the price of Firm A's product. The displayed independent variables are indicators for treatment groups defined in Table III. We regress normalized willingness to pay on indicator variables for the six treatment groups, omitting the indicator for the control group. We estimate a separate regression for each industry. We estimate a participant-question-level regression. Standard errors, clustered at the participant level, are shown in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

	Normalized WTP		
	(1)	(2)	(3)
Bankruptcy	-0.218*** (0.021)	-0.179*** (0.021)	-0.193*** (0.022)
Prebankruptcy	-0.196*** (0.024)	-0.101*** (0.026)	-0.173*** (0.025)
Postbankruptcy	-0.076*** (0.019)	-0.070*** (0.023)	-0.048** (0.024)
Quality	-0.081*** (0.021)	-0.069*** (0.024)	-0.080*** (0.025)
Survival 50	-0.310*** (0.026)	-0.123*** (0.027)	-0.208*** (0.025)
Survival 100	-0.146*** (0.022)	-0.114*** (0.023)	-0.074*** (0.027)
Constant	0.898*** (0.011)	0.962*** (0.015)	0.879*** (0.015)
Industry	Airline	Retail	Car
Observations	6,996	5,247	5,247

6.4 Table IV: How, When, and Why Bankruptcy Changes Willingness to Pay

Read: The bankruptcy coefficient is -0.218 for airlines, -0.179 for retail, and -0.193 for car manufacturers. Prebankruptcy information produces nearly as large an effect, but the paper later shows consumers are usually unaware of prebankruptcy distress. Postbankruptcy effects are smaller. Quality reassurance attenuates much of the effect, and survival reassurance also matters.

Interpretation: The table identifies mechanisms. Current quality concerns account for much of the effect in airlines and retail, while future interactions are especially important for cars.

Economic message: Consumers avoid bankrupt firms because bankruptcy threatens product/service quality and future relationship value, not mainly because it is a permanent quality signal.

6.5 Table V: Awareness, Bankruptcy Reactions, and Firm Characteristics

Read: A previous filing raises perceived bankruptcy awareness by 14.2 percentage points. Public and consumer-facing firms are more recognized. Among consumer-facing bankrupt firms, average awareness is 37.8% and average reduce-WTP response is 43.9%.

Interpretation: The table shows that bankruptcy awareness is widespread and economically relevant outside the experiment.

Economic message: Demand effects matter because a large share of the customer base actually knows when major consumer-facing firms file.

Table V

Awareness, Bankruptcy Reactions, and Firm Characteristics

For 99 firms, we calculate the fraction of adults who believe each firm has filed for bankruptcy and the fraction who would reduce their willingness to pay if it were to file. We estimate firm-level OLS regressions using these fractions as outcome variables. In column (1), we include 70 previously bankrupt firms and 29 never-bankrupt firms. In column (2), we include only 70 previously bankrupt firms. In columns (3) and (4), we include only consumer-facing previously bankrupt firms. See Section III.C for details. In each column, we report the mean of the dependent variable. We report robust standard errors in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

	Aware of Bankruptcy			Reduce WTP
	(1)	(2)	(3)	(4)
Bankrupt	0.142*** (0.028)			
Liabilities		-0.000 (0.000)		
Public		0.075** (0.036)		
Consumer facing		0.179*** (0.033)		
Press coverage		-0.000 (0.000)		
Apparel			0.255*** (0.042)	0.320*** (0.081)
Automotive			0.388*** (0.065)	0.454*** (0.021)
Aviation			0.350*** (0.042)	0.492*** (0.036)
Entertainment			0.385*** (0.069)	0.452*** (0.026)
Hotel			0.534*** (0.131)	0.568*** (0.052)
Manufacturing			0.347*** (0.097)	0.357*** (0.043)
Retail			0.437*** (0.050)	0.413*** (0.022)
Telecommunications			0.171*** (0.047)	0.497*** (0.035)
Constant	0.187*** (0.014)	0.137*** (0.030)		
Dependent variable mean	0.287	0.329	0.378	0.439
Sample	All Firms	All Bankrupt	Consumer Facing	Consumer Facing
Observations	99	70	53	53

Table VI

Consumer Ignorance of Prebankruptcy Distress

In our mechanism experiment (Section II), we ask participants to rate 25 firms based on how close the firms came to bankruptcy over the period 2010 to 2019. Participants report this measure, *Near-Bankruptcy Awareness*, on a scale from 1 (never close) to 5 (very close). For each firm, we define *Actual Bankruptcy* as an indicator equal to one if the firm has ever filed for bankruptcy. We regress *Near-Bankruptcy Awareness* on *Actual Bankruptcy* in a firm-level data set and report the result in column (1). Next, we calculate *Worst Credit Rating*, the worst credit rating that each firm received between 2010 and 2019, coded on a numerical scale from 1 (AAA) to 22 (D). We assign a value of zero to firms with no debt. We regress *Near-Bankruptcy Awareness* on both *Actual Bankruptcy* and *Worst Credit Rating*, reporting the results in column (2). Finally, we exclude firms that ever filed for bankruptcy and regress *Near-Bankruptcy Awareness* on *Worst Credit Rating*, reporting the result in column (3). We report robust standard errors in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

	Near-Bankruptcy Awareness		
	(1)	(2)	(3)
Actual bankruptcy	0.404** (0.157)	0.330** (0.153)	
Worst credit rating		0.010 (0.012)	0.010 (0.013)
Constant	2.372*** (0.071)	2.275*** (0.156)	2.272*** (0.163)
Observations	25	25	18
Adj. R ²	0.225	0.213	-0.0176

6.6 Table VI: Consumer Ignorance of Prebankruptcy Distress

Read: Consumers rate actual bankrupt firms as much closer to bankruptcy, but among firms that did not file, ratings are not correlated with worst credit rating. Conditional on actual bankruptcy, credit rating does not predict perceived near-bankruptcy status.

Interpretation: Consumers learn the bankruptcy filing but do not track prebankruptcy financial distress.

Economic message: Indirect costs are unlikely to fully arise before filing; Chapter 11 itself reveals or publicizes the distress.

6.7 Table VII: Awareness, Bankruptcy Reactions, and Participant Characteristics

Read: Prior customers are 24.2 percentage points more likely to know about a firm's bankruptcy. They are also less likely to report reducing WTP in the hypothetical survey. Conditional on observables, awareness itself has virtually no relation with reduce-WTP.

Interpretation: Awareness is not random; customers know more. But once observables are controlled for, aware and unaware consumers have similar aversion to bankruptcy.

Economic message: Experimental estimates based on newly informed consumers can plausibly generalize to aware consumers once customer composition is considered.

Table VII

Awareness, Bankruptcy Reactions, and Participant Characteristics

We survey participants about which firms have filed for bankruptcy. Focusing on 53 previously bankrupt consumer-facing firms, we construct a participant-firm level data set with two outcome variables: an indicator equal to one if the participant is aware of the firm's bankruptcy and an indicator equal to one if the participant would reduce their willingness to pay if the firm were to hypothetically file for bankruptcy. In column (1), we regress awareness on an indicator for the participant previously purchasing from the firm and participant-level variables capturing the participant's demographics and ignorance about bankruptcy. In column (3), we estimate the same regression using the reduced-willingness-to-pay outcome variable and using awareness as an independent variable. In column (2), we simply regress the reduced-willingness-to-pay outcome on awareness. See Section III E for details. Standard errors, clustered at the participant level, are shown in parentheses. For each demographic characteristic (i.e., income), we use a Wald test to calculate a joint p -value for all of the corresponding indicator variables. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

	Aware of Bankruptcy	Reduce WTP	
	(1)	(2)	(3)
Aware of bankruptcy		-0.050*	0.012 (0.026)
Prior customer of the firm	0.242*** (0.023)		-0.170*** (0.025)
Bankruptcy-process ignorance	-0.001 (0.001)		0.005*** (0.001)
Constant		0.458*** (0.023)	
p -Value for education indicators	0.617		0.312
p -Value for income indicators	0.0400		0.328
p -Value for age indicators	0.527		0
p -Value for male indicator	0.626		0.311
p -Value for ethnicity indicators	0.173		0.00900
p -Value for country-of-birth indicators	0		0
Control variables	Y	N	Y
Observations	2,228	2,234	2,228

Table VIII

Heterogeneous Price-List-Experiment Treatment Effects

For each of our three price list experiments, we predict bankruptcy awareness using observable participant characteristics. Specifically, using control-group participants from one experiment at a time, we regress the awareness indicator $Aware_i$ on the vector X_i of control variables described in Section I (e.g., participant demographics and purchase histories). For each regression, we use the estimated coefficients and the vector X_i to construct fitted values, $Aware_i^{pred}$, for both control-group and treatment-group participants. In each experiment sample, we calculate the median of $Aware_i^{pred}$ and partition the sample into above-median values of $Aware_i^{pred}$ and below-median values of $Aware_i^{pred}$. For each subsample of each experiment, we estimate the 2SLS regression of WTP_i on $Aware_i$ (instrumented by our treatment) described in Section I. Columns (1), (3), and (5) display 2SLS estimates for samples with below-median predicted awareness from the Hertz, Instant Pot, and Spirit experiments, respectively. Columns (2), (4), and (6) display analogous estimates for participants with above-median predicted awareness. For each subsample of each experiment sample, we display the average realized awareness $Aware_i$ for control-group participants within that subsample. We report robust standard errors in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

	WTP					
	(1)	(2)	(3)	(4)	(5)	(6)
Aware of bankruptcy	-10.150** (4.766)	-14.095** (5.846)	-16.711** (7.077)	4.035 (7.448)	-6.920** (2.961)	-11.243*** (3.861)
Firm	Hertz	Hertz	Instant	Instant	Spirit	Spirit
Predicted awareness	Low	High	Low	High	Low	High
Control variables	Y	Y	Y	Y	Y	Y
Control awareness	0.100	0.340	0	0.0500	0.230	0.430
Observations	178	178	81	80	776	775

6.8 Table VIII: Heterogeneous Price-List-Experiment Treatment Effects

Read: Likely-aware participants respond strongly in the Hertz and Spirit experiments, with point estimates sometimes larger than for likely-unaware participants. The Instant Pot high-awareness subsample is imprecise.

Interpretation: The table checks whether compliers in the main 2SLS are special. Overall, the results support external validity to consumers who are naturally aware of bankruptcies.

Economic message: Bankruptcy aversion is not limited to consumers who learn about bankruptcy experimentally.

6.9 Table IX: Model-Implied Bankruptcy Impacts

Read: The model implies immediate market-share declines of 34.1% for Delta, 27% for American, and 28% for United. Current indirect costs equal 37.2%, 29%, and 29.8% of current profits. Present-value indirect costs are 15.1%, 11.7%, and 14.6% of future profits.

Interpretation: This table converts consumer WTP evidence into firm-value terms. The immediate price response is small, so the main channel is lost demand rather than firms cutting prices enough to offset the demand shock.

Economic message: Customer attrition alone creates indirect bankruptcy costs on the same order as many estimates of total distress costs in the corporate-finance literature.

Table IX
Model-Implied Bankruptcy Impacts

Using the estimated model of Section V, we calculate the impacts of historical bankruptcies. Each column corresponds to a major historical airline bankruptcy: Delta Air Lines, American Airlines, and United Airlines. The first row reports the immediate percent decline in market share that each airline experienced because of its bankruptcy. The second row reports the percent change in price each firm chose in response to its bankruptcy, calculated as a quantity-weighted average across routes. The third row reports the ratio of the immediate indirect cost (profit decline) that each firm experienced to the profit the firm would have received if it had not filed for bankruptcy. The fourth row reports the immediate percent change in consumer surplus, across all routes and all customers, that occurred because of the bankruptcy. The fifth row reports the largest percent change in price chosen by a competitor of the bankrupt firm in response to the bankruptcy, calculated as a quantity-weighted average across routes. Finally, the sixth row reports the ratio of the present value of all future indirect costs (lost profits) experienced by the bankrupt firm to the present value of all of the firm's future profits. See Section VC for details.

	Delta	American	United
Market share	-34.1	-27	-28
Own-price change	-0.3	-0.2	-0.2
Current indirect cost/Current profit	-37.2	-29	-29.8
Consumer surplus	-3.4	-2.6	-2.5
Max competitor-price change	0.2	0.1	0.1
PV(Indirect costs)/PV(Profits)	-15.1	-11.7	-14.6

Table IX: Model-Implied Bankruptcy Impacts.

7 Interpretive synthesis

- **Experimental foundation.** Random information provision identifies the causal WTP penalty from the bankruptcy label.
- **Awareness margin.** Many consumers know about actual major bankruptcies, but not about pre-bankruptcy distress.
- **Mechanism.** The WTP penalty comes from current-quality concerns and future-interaction concerns.
- **External validity.** Awareness and heterogeneity analyses suggest effects are not limited to laboratory compliers.
- **Structural translation.** Airline model estimates show that WTP penalties translate into substantial market-share and profit losses.

8 Short summary

- The paper estimates indirect bankruptcy costs from lost customers.
- Randomized price-list experiments show that learning about Chapter 11 reduces WTP by 17% to 28%.
- Mechanism experiments show that consumers worry about current quality and future interactions.
- Consumers are broadly aware of actual major bankruptcies but not prebankruptcy distress.
- Structural airline estimates imply indirect costs of 12% to 15% of enterprise value.
- The incremental contribution is to combine randomized WTP experiments, bankruptcy-awareness measurement, mechanism decomposition, and structural IO cost quantification.

9 Conclusion

9.1 Core conclusion

Corporate bankruptcy filings causally reduce consumer demand for bankrupt firms' products. The resulting lost-customer costs are large enough to matter for capital structure and restructuring decisions.

9.2 Economic intuition

Consumers avoid bankrupt firms because they fear lower current quality and the loss of future relationship value. These concerns are especially important when products involve warranties, loyalty programs, service continuity, or durable relationships.

9.3 Incremental contribution in one sentence

The paper provides the first experimental and structural quantification of consumer-driven indirect bankruptcy costs and shows that customer attrition can account for a large share of firm value losses in Chapter 11.

9.4 Limitations

The structural estimates omit benefits of bankruptcy and do not model all indirect costs, such as employee attrition. They should be read as estimates of customer-driven costs conditional on distress that necessitates filing, not as net welfare effects of bankruptcy law.

9.5 Takeaway

Bankruptcy is not only a legal and financial event. It is also a consumer-information event that changes demand, market shares, and firm value.